

LIBRARY MANAGEMENT SYSTEM

Submitted By:

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Abstract

The Library Management System is a backend application developed using FastAPI and MongoDB. It manages students, books, borrowing operations, staff and departments. The system supports CRUD operations, relationship mapping, due-date tracking and API testing through Swagger.

Objective

- Develop a Library Management System using MongoDB
- Implement CRUD operations
- Demonstrate document database relationships
- Provide API interaction using FastAPI

Technologies Used

Python
FastAPI
MongoDB
PyMongo
Uvicorn
Swagger

4. System Architecture

User → Swagger → FastAPI → MongoDB

6. Database Design

Students(_id, name, borrowedBooks)
Books(_id, title, author, available)
Departments(_id, name)
Staff(_id, name, departmentId)

7. Features Implemented

Student CRUD
Book CRUD
Borrow/Return
Due Dates
Book Availability
Search Books
Student Status
Staff and Departments

8. Result

Project successfully performed CRUD operations and demonstrated MongoDB relationships using FastAPI APIs.

9. Conclusion

The project demonstrates backend development using FastAPI with MongoDB and provides scalable document-based data management.

10. Future Enhancements

Authentication
Email Notifications
Fine Calculation
Frontend Dashboard

Screenshots

Library Management System 1.0 OAS 3.1

/openapi.json

FastAPI + MongoDB CRUD Project

default

GET	/	Home	⌵
POST	/students	Create Student	⌵
GET	/students	Get Students	⌵
PUT	/students/{id}	Update Student	⌵
DELETE	/students/{id}	Delete Student	⌵
POST	/books	Create Book	⌵
GET	/books	Get Books	⌵
PUT	/books/{id}	Update Book	⌵
DELETE	/books/{id}	Delete Book	⌵
PUT	/borrow	Borrow Book	⌵
GET	/departments	Get Departments	⌵
POST	/departments	Create Department	⌵
POST	/staff	Create Staff	⌵
GET	/staff	Get Staff	⌵
PUT	/return	Return Book	⌵
GET	/books/search	Search Books	⌵
GET	/students/status	Student Status	⌵

FastAPI Swagger Home

POST

/students

Create Student

⌵

Parameters

Cancel

Name	Description
name required	
string	
(query)	

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/students/name-PrakashRaj' \
  -H 'accept: application/json' \
  -d ''
```

Request URL

```
http://127.0.0.1:8080/students/name-PrakashRaj
```

Server response

Code	Details
200	Response body <pre>{ "message": "Student added" }</pre> Download Response headers <pre>content-length: 27 content-type: application/json date: Thu, 18 Jun 2020 01:02:22 GMT server: uSLoRn</pre>

Response

Code	Description	Links
200	Successful Response Media type application/json Content Accept header Example Value Schema <pre>"string"</pre>	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", ""], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }</pre>	No links

Get /students

Get Students

Parameters

Cancel

No parameters

Execute

Clear

Response

Curl

curl -X "GET" \n'http://127.0.0.1:3000/students' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:3000/students

Server response

Code

Details

200

Response body

```
{\n  "_id": "64166c72863a06f6d7dc1",\n  "name": "Vishal Sharma Prakash",\n  "bornandbook": [\n    {\n      "bookId": "64166c72863a06f6d7dc1",\n      "date": "2024-06-01",\n    },\n    {\n      "bookId": "64166c72863a06f6d7dc1",\n      "date": "2024-07-01",\n    }\n  ],\n  "_id": "64166c72863a06f6d7dc2",\n  "name": "Arjun Kumar",\n  "_id": "64166c72863a06f6d7dc3",\n  "name": "Vijay Nair",\n  "_id": "64166c72863a06f6d7dc4",\n  "name": "Prakash Raj",\n}
```

Response headers

content-length: 378\ncontent-type: application/json\ndate: Thu, 18 Jun 2020 11:14:07 GMT\nserver: nodejs

Response

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Control accept header

Example Value

Schema

"string"

PUT

/students/{id} Update Student

Cancel

Name	Description
id required string (path)	6a33f36702ab8cb20c6a253
name required string (query)	Prakash Bedi

Execute

Clear

Responses

curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8080/students/6a33f36702ab8cb20c6a253?name=PrakashBedi' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/students/6a33f36702ab8cb20c6a253?name=PrakashBedi
```

Server response

Code	Details
200	Response body <pre>{ "message": "updated" }</pre> Download Response headers <pre>content-length: 21 content-type: application/json date: Thu, 28 Jun 2024 11:08:17 GMT server: nginx</pre>

Response

Code	Description	Links
200	Successful Response Media type application/json Content Accept header Example Value Schema <pre>"string"</pre>	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", "msg": "string", "type": "string", "value": "string", "ctx": {}] }] }</pre>	No links

DELETE

/students/{id} Delete Student

Parameters

Cancel

Name	Description
id required	
string (path)	6a33f36702ab9cbc20c6a263

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8080/students/6a33f36702ab9cbc20c6a263' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/students/6a33f36702ab9cbc20c6a263
```

Server response

Code	Detail
200	Response body <pre>{ "message": "Student Deleted" }</pre> Download Response headers <pre>content-length: 28 content-type: application/json date: Thu, 18 Jun 2020 13:06:38 GMT server: unicorn</pre>

Response

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header Example Value Schema <pre>"string"</pre>	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string",], "msg": "string", "path": "string", "type": "string", "ctx": {} }] }</pre>	No links

POST /students execution

POST

/books Create Book

Parameters

Cancel

Name	Description
title required	
string (query)	MongoDB: The Definitive Guide
author required	
string (query)	Kristina Chodorow

ExecuteClear

Response

Curl

curl -X 'POST' \n'http://127.0.0.1:3000/books?title=MongoDB%20The%20Definitive%20Guide&author=KristinaChodorow' \n-H 'accept: application/json' \n-d ''

Request URL

http://127.0.0.1:3000/books?title=MongoDB%20The%20Definitive%20Guide&author=KristinaChodorow

Server response

Code	Detail
200	Response body {\n "message": "book added"\n} Response headers content-length: 34\ncontent-type: application/json\ndate: Thu, 18 Jun 2020 11:37:46 GMT\nserver: node

Response

Code	Description	Link
200	Successful Response	No link
Media type: application/json		
Controls: accept header		
Example Value Schema		
"string"		
422	Validation Error	No link
Media type: application/json		
Example Value Schema		
{\n "detail": [\n {\n "loc": [\n "string",\n "\n],\n "msg": "string",\n "type": "string",\n "value": "string",\n "ctx": {}\n }\n]\n}		

PUT

/books/{id} Update Book

Parameters

Cancel

Name	Description
id required string (path)	6a33f4a802ab5cbc20c6a254
title required string (query)	MongoDB: The Indefinive Guide

Execute

Clear

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:3000/books/6a33f4a802ab5cbc20c6a254?title=MongoDB: The Indefinive Guide' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:3000/books/6a33f4a802ab5cbc20c6a254?title=MongoDB: The Indefinive Guide
```

Server response

Code	Detail
200	Response body <pre>{ "message": "Book updated" }</pre> Download Response headers <pre>content-length: 28 content-type: application/json date: Thu, 08 Jun 2023 13:06:07 GMT server: express</pre>

Response

Code	Description	Links
200	Successful Response Media type application/json Control Accept header Example Value Schema <pre>"string"</pre>	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", "string", "type": "string", "input": "string", "ctx": {}] }] }</pre>	No links

Delete Book

/books/{id}

Parameters

Name	Description
id required	
string	6a33f4a802ab5cbc20c6a254
(path)	

Execute

Clear

Response

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8080/books/6a33f4a802ab5cbc20c6a254' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8080/books/6a33f4a802ab5cbc20c6a254

Server response

Code	Detail
200	Response body <pre>{ "message": "Book deleted" }</pre> Response headers <pre>content-length: 36 content-type: application/json date: Thu, 12 Jun 2020 13:41:46 GMT server: node</pre>

Responses

Code	Description	Link
200	Successful Response Media type application/json Control: Accept header Example Value Schema <pre>"string"</pre>	No link
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", ""], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }</pre>	No link

GET /books/search

PUT /borrow Borrow Book

Cancel

Name	Description
student_id required string (query)	6a33afc4a009353f6b5b10e0
book_id required string (query)	6a33aef3a009353f6b5b10dc

Execute
Clear

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:3000/borrow?student_id=6a33afc4a009353f6b5b10e0&book_id=6a33aef3a009353f6b5b10dc' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:3000/borrow?student_id=6a33afc4a009353f6b5b10e0&book_id=6a33aef3a009353f6b5b10dc
```

Server response

Code	Details
200	Response body <pre>{ "message": "Book borrowed", "dueDate": "2024-07-02" }</pre> Download Response headers <pre>content-length: 58 content-type: application/json date: Thu, 18 Jun 2020 22:03:04 GMT server: nodecore</pre>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls: Accept header Example Value Schema <pre>"string"</pre>	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", "string", "type": "string", "value": "string", "ctx": {}] }] }</pre>	No links

PUT /borrow with due date

GET /students/status Student Status

Parameters

Cancel

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/students/status' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/students/status
```

Server response

CodeDetails

200

Response body

```
{
  "student": "Vishal Dharsan Prakash",
  "status": "OVERDUE"
},
{
  "student": "Arthur Morgan",
  "status": "CLEAR"
},
{
  "student": "Vijay Hari",
  "status": "CLEAR"
}
]
```

Response headers

```
content-length: 145
content-type: application/json
date: Thu, 18 Jun 2020 13:45:43 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header Example Value Schema<pre>"string"</pre>	No links

GET /students/status

```
_id: ObjectId('6a31454729835ad66fdd7db1')
name: "Vishaal Dharsan Prakash"
▼ borrowedBooks: Array (2)
  ▼ 0: Object
    bookId: ObjectId('6a314a5a743587b658c45f58')
    dueDate: "2026-06-02"
  ▼ 1: Object
    bookId: ObjectId('6a33af0aa009353f6b5b10dd')
    dueDate: "2026-07-02"
```

```
_id: ObjectId('6a33af53a009353f6b5b10df')
name: "Arthur Morgan"
```

```
_id: ObjectId('6a33afc4a009353f6b5b10e0')
name: "Vijay Hari"
▶ borrowedBooks: Array (1)
```

```
_id: ObjectId('6a314a5a743587b658c45f58')
title: "The Blood Meridian"
author: "Cormac McCarthy"
available: false
```

```
_id: ObjectId('6a33abfd7444ac560913045e')
title: "Atomic Habits"
author: "James Clear"
available: true
```

```
_id: ObjectId('6a33aef3a009353f6b5b10dc')
title: "The Great Gatsby"
author: "F Scott Fitzgerald"
available: false
```

```
_id: ObjectId('6a33af0aa009353f6b5b10dd')
title: "The Godfather"
author: "Mario Puzo"
available: false
```

```
_id: ObjectId('6a33af25a009353f6b5b10de')
title: "Harry Potter"
author: "JK Rowling"
available: true
```

```
_id: ObjectId('6a33add8a009353f6b5b10d5')  
name : "Administration"
```

```
_id: ObjectId('6a33adeca009353f6b5b10d6')  
name : "Cataloging"
```

```
_id: ObjectId('6a33adf4a009353f6b5b10d7')  
name : "Circulation"
```

```
_id: ObjectId('6a33ae30a009353f6b5b10d8')  
name : "Samuel "  
departmentId : ObjectId('6a33add8a009353f6b5b10d5')
```

```
_id: ObjectId('6a33ae5fa009353f6b5b10d9')  
name : "Unnikrishnan"  
departmentId : ObjectId('6a33add8a009353f6b5b10d5')
```

```
_id: ObjectId('6a33ae82a009353f6b5b10da')  
name : "Ryan "  
departmentId : ObjectId('6a33adeca009353f6b5b10d6')
```

```
_id: ObjectId('6a33aea7a009353f6b5b10db')  
name : "Syed"  
departmentId : ObjectId('6a33adf4a009353f6b5b10d7')
```

MongoDB Compass collections

```

[notice] A new release of pip is available: 24.0 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip

(venv) C:\Users\visha\OneDrive\Desktop\MONGODB TASK\LIBRARY_MANAGER_PROJECT>uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\visha\\OneDrive\\Desktop\\MONGODB TASK\\LIBRARY_MANAGER_PROJECT']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [15408] using StatReload
INFO: Started server process [14992]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:54118 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:54118 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:55211 - "POST /students?name=Prakash%20Raj HTTP/1.1" 200 OK
INFO: 127.0.0.1:56962 - "GET /students HTTP/1.1" 200 OK
INFO: 127.0.0.1:58904 - "PUT /students/6a33f36702ab9cbc20c6a253?name=Prakash%20Bedi HTTP/1.1" 200 OK
INFO: 127.0.0.1:56783 - "DELETE /students/6a33f36702ab9cbc20c6a253 HTTP/1.1" 200 OK
INFO: 127.0.0.1:65513 - "POST /books?title=MongoDB%3A%20The%20Definitive%20Guide&author=Kristina%20Chodorow%20 HTTP/1.1" 200 OK
INFO: 127.0.0.1:54465 - "GET /books HTTP/1.1" 200 OK
INFO: 127.0.0.1:62621 - "PUT /books/6a33f4a802ab9cbc20c6a254?title=MangoDB%3A%20The%20Indefinitive%20Guide HTTP/1.1" 200 OK
INFO: 127.0.0.1:54752 - "DELETE /books/6a33f4a802ab9cbc20c6a254 HTTP/1.1" 200 OK
INFO: 127.0.0.1:56451 - "PUT /borrow?student_id=6a33afc4a009353f6b5b10e0&book_id=6a33aef3a009353f6b5b10dc HTTP/1.1" 200 OK
INFO: 127.0.0.1:53338 - "GET /students/status HTTP/1.1" 200 OK

```

FastAPI terminal running